Do Patient Metadata Still Help Strong Image Models? Capacity-Dependent Attenuation of Auxiliary Information in Skin Lesion Classification

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Abstract—Fusing patient metadata with lesion images is widely reported to improve skin cancer classification. We ask whether this benefit is a property of the data or of a weak image model. We define the added value of metadata as the out-of-fold reduction in cross-entropy when metadata are added to an image score, and show that its population counterpart, $I(Y; A \mid S)$, decomposes exactly into the information the image cannot carry, $I(Y; A \mid X)$, plus a residual that vanishes as the representation S approaches the image X. We measure added value along a ladder of five frozen image representations, from colour histograms to a vision transformer, with grouped cross-validation and grouped bootstrap inference. In a pre-registered analysis on HAM10000, the added value of metadata for melanoma detection falls from 18.6×10^{-3} to 1.7×10^{-3} nats from colour histograms to ConvNeXt-T (AUROC gain 0.043 to 0.003), and the fall survives adjustment for image source, a strong shortcut. In a pre-registered replication, attenuation from colour histograms to ConvNeXt-T held after Holm correction in 3 of 3 independent datasets, and attenuation within deep representations in 1 of 3. The share of the metadata's standalone information that ConvNeXt-T leaves non-redundant varies widely between datasets, from 0.07 (HAM10000) to 0.50 (PAD-UFES-20). Attenuation also holds in a complete-case analysis that removes label-informative missingness, which inflated PAD-UFES-20's metadata signal. An exact spatial permutation test shows that classifier evidence concentrates on expert-segmented lesions, that missed melanomas are smaller rather than mislocalised, and that naive pixel permutation roughly doubles the number of significant maps.

Index Terms—skin lesion classification, patient metadata, multimodal fusion, conditional mutual information, model capacity, interpretability, spatial permutation test

I. Introduction

Clinicians do not diagnose skin lesions from images alone: age, anatomical site, and lesion history all shape the assessment. Following this intuition, many automated systems fuse patient metadata with image features. A systematic review of such systems found that every included study reported a gain from patient data, while noting that it remains unclear how each variable contributes [1]. On the smartphone dataset PAD-UFES-20, adding clinical data improved balanced accuracy by about 7% across several networks [2].

A reported gain, however, conflates two things: information in the metadata that the image genuinely lacks, and information the image contains but a particular model failed to extract. A patient's age, for example, is partly visible in the skin around a lesion. If an image model is weak, metadata can compensate for what it misses; if it is strong, the same metadata may be redundant. Whether metadata "help" may therefore depend on the image model they are added to, a dependence that single-backbone studies cannot reveal.

This paper makes four contributions:

1) An interpretation of added value. Using the chain rule of mutual information, we write the Bayes added value of metadata A to a representation $S = g(X)$ of an image X as $I(Y; A \mid S) = I(Y; A \mid X) + R(S)$, where $I(Y; A \mid X)$ is the information unavailable from the image and the residual $R(S)$ vanishes as $S \to X$ (Proposition 1). The decomposition is elementary; its value is that it says what a measured gain from metadata does and does not show.
2) A measurement protocol. We estimate added value out of fold along a ladder of frozen image representations, with grouped cross-validation, inner-fold tuning, and grouped bootstrap inference, and we audit metadata for label leakage.
3) Evidence across datasets. A pre-registered analysis on HAM10000 [3] and a pre-registered replication on ISIC 2020 [4], ISIC 2019 excluding HAM10000 [5], and PAD-UFES-20 [6], together with a pre-registered mechanism test whose failure is itself informative: symptoms assumed invisible in photographs turned out to be partly visible.
4) A calibrated interpretability test. We compare class-activation evidence with expert lesion masks using an exact spatial permutation test computed by FFT, and we document two pitfalls: invalid pixel permutation and a sign convention that manufactures an apparent link between errors and off-lesion attention.

II. Related Work

Metadata fusion in dermatology. Concatenating one-hot metadata with image features is the dominant approach, and all eleven studies in a systematic review reported a

gain [1]. To our knowledge no study has measured added value along a controlled ladder of image models with grouped inference.

Incremental value. In clinical prediction, the added value of a new variable is commonly tested with nested models; comparing nested-model AUCs with DeLong's test is invalid [7]. In logistic models, omitting a predictor that is independent of a covariate still attenuates that covariate's coefficient (non-collapsibility) [8]. This mechanism makes directly acting metadata gain value as the image model improves (Section III).

Leakage and shortcuts. Splitting medical images without grouping by subject inflates accuracy substantially [9]. Dermatology datasets contain several images per lesion and per patient, and acquisition source can predict diagnosis.

Spatial nulls. Comparisons between spatial maps must preserve spatial autocorrelation in the null distribution; naive voxel permutation is anti-conservative [10], [11]. We carry this principle over to the evaluation of saliency maps against anatomy.

III. Theory

Let X be an image, A metadata, $Y \in \{0,1\}$ the label, and $S = g(X)$ a representation. With log-loss $\ell(y,p) = -\log p(y)$, the Bayes predictors satisfy $\mathbb{E}\,\ell(Y, P(Y \mid S)) = H(Y \mid S)$, so the expected reduction in log-loss from adding A is

$$\Delta(S) = H(Y \mid S) - H(Y \mid S, A) = I(Y; A \mid S). \tag{1}$$

Proposition 1. For any measurable g and $S = g(X)$,

$$I(Y; A \mid S) = I(Y; A \mid X) + \underbrace{I(Y; X \mid S) - I(Y; X \mid S, A)}_{R(S)}. \tag{2}$$

If g is injective, $R(S) = 0$. If moreover $Y \perp\!\!\!\perp A \mid X$, then $\Delta(S) = 0$.

Proof. Expand $I(Y; A, X \mid S)$ with the chain rule in two orders [12]: $I(Y; A \mid S) + I(Y; X \mid S, A) = I(Y; X \mid S) + I(Y; A \mid X, S)$. Because S is a function of X, conditioning on (X, S) equals conditioning on X, so $I(Y; A \mid X, S) = I(Y; A \mid X)$. Rearranging gives (2). If g is injective, X is a function of S, so both terms of $R(S)$ vanish. The final claim follows because $I(Y; A \mid X) = 0$ when $Y \perp\!\!\!\perp A \mid X$. □

Interpretation. $I(Y; A \mid X)$ is the part of the metadata's information that no image model can recover. $R(S)$ measures how the representation's losses interact with A. $R(S) > 0$ when A substitutes for image information that S discarded; added value then attenuates as S improves. $R(S) < 0$ indicates synergy; added value then grows as S improves, which is the information-theoretic face of non-collapsibility [8]. Proposition 1 was checked numerically to machine precision on 200 random discrete distributions and coarsenings (code release).

Estimation. We estimate Δ for each rung k of a capacity ladder by a two-stage out-of-fold procedure. First, an image score $s_k(x)$ is obtained from frozen features $f_k(x)$ by L_2-regularised logistic regression, with the penalty chosen by grouped inner cross-validation within each training fold. Second, logistic models on $[s_k]$ and $[s_k, a]$ are fitted on the same outer folds, and

$$\hat{\Delta}_k = \frac{1}{n}\sum_{i=1}^{n}\Big[\ell\big(y_i, \hat{p}_0^{(-i)}\big) - \ell\big(y_i, \hat{p}_1^{(-i)}\big)\Big], \tag{3}$$

where $(-i)$ denotes a model fitted without the fold containing i. Because both models are estimated, $\hat{\Delta}_k$ targets the reduction achievable by the second-stage model family, not $I(Y; A \mid S_k)$ exactly, and it can be negative in finite samples. Uncertainty is quantified by a bootstrap over grouping units (lesions or patients), and differences between rungs by a paired grouped bootstrap of per-image loss differences.

Interpretability test. For a linear head w on global-average-pooled features $f = \frac{1}{|\Omega|}\sum_{u\in\Omega} F(u)$, the class activation map $c(u) = \sum_j w_j F_j(u)$ [13] decomposes the logit exactly. For a non-negative evidence map e and a binary mask m on a periodic grid Ω, the mass inside the mask after a shift by τ is

$$M(\tau) = \frac{\sum_u e(u)\, m(u-\tau)}{\sum_u e(u)} = \frac{(e \star m)(\tau)}{\|e\|_1}, \tag{4}$$

a circular cross-correlation. Hence $\mathcal{F}^{-1}[\mathcal{F}(e)\overline{\mathcal{F}(m)}]$ yields $M(\tau)$ for all $|\Omega|$ shifts in $O(|\Omega|\log|\Omega|)$, and the exact p-value is $|\{\tau : M(\tau) \geq M(0)\}|/|\Omega|$. Unlike pixel permutation, the shift null preserves the spatial autocorrelation of e.

IV. Experimental Setup

Datasets (Table I). HAM10000 (dermoscopy; 10,015 images of 7,470 lesions) [3]; ISIC 2020 (dermoscopy; 33,126 images of 2,056 patients, 584 melanomas) [4]; ISIC 2019 with all HAM10000 lesions removed [5]; and PAD-UFES-20 (smartphone clinical images; 2,298 images of 1,373 patients) [6].

Leakage audit. Fields that encode how the diagnosis was established were excluded: HAM10000's dx_type (alone, AUROC 0.76 for melanoma) and PAD-UFES-20's biopsed (every cancer is biopsied). On HAM10000 the image source alone predicts melanoma (AUROC 0.73), and frozen deep features identify the source almost perfectly (AUROC up to 0.995), so we report source-adjusted analyses. A separate audit of missingness found that it, too, can carry the label. In PAD-UFES-20, all 804 images lacking gender, skin type, and history fields are benign and unbiopsied, so missingness is a proxy for the excluded biopsed field, and "unknown" growth and change histories are 88% malignant. Smaller associations exist elsewhere (e.g. ISIC 2020: no melanoma among the 65 images of unknown sex). Because the pre-registered encoding keeps missing-value

TABLE I

Datasets. Positives: melanoma (malignant for PAD-UFES-20). Groups define the cross-validation and bootstrap units. *Images without a lesion identifier form their own group.

Dataset	Modality	Images	Groups	Pos.	Metadata
HAM10000	dermoscopy	10,015	7,470 lesions	1,113	age, sex, site
ISIC 2020	dermoscopy	33,126	2,056 patients	584	age, sex, site
ISIC 2019 (non-HAM)	dermoscopy	15,316	6,461 lesions*	3,409	age, sex, site
PAD-UFES-20	smartphone	2,298	1,373 patients	1,089	demogr. + symptoms

TABLE II

HAM10000, melanoma vs. rest, lesion-grouped 5-fold CV. AUROC and $\hat{\Delta}$ are means over three fold assignments; the CI is the 95% lesion bootstrap for assignment 0. Recoverability: out-of-fold R^2 for age and mean AUROC for the five commonest sites and for image source, predicted from the frozen features.

Representation	Image AUROC	$\hat{\Delta}$ (10^{-3} nats)	95% CI (10^{-3})	Recoverability age	site	source
HSV histogram	0.765	18.6	[12.2, 24.3]	0.06	0.65	0.921
ResNet-18	0.853	3.7	[-0.1, 8.1]	0.28	0.73	0.979
ResNet-50	0.850	3.0	[-0.5, 6.9]	0.32	0.73	0.989
ConvNeXt-T	0.889	1.7	[-2.7, 5.6]	0.39	0.79	0.995
ViT-B/16	0.887	2.2	[-1.1, 6.3]	0.37	0.79	0.995

TABLE III

HAM10000 robustness (fold assignment 0). ΔAUROC and ΔBrier (10^{-3}) from adding metadata, with 95% lesion-bootstrap CIs. Last two columns: out-of-fold log-loss of score + metadata (two-stage, as in Table II) vs. all frozen features + metadata in one model; similar values mean the one-dimensional score loses little.

Representation	ΔAUROC [95% CI]	ΔBrier [95% CI]	Log-loss with metadata two-stage	joint
HSV histogram	+0.043 [+0.030, +0.056]	4.84 [3.02, 6.70]	0.288	0.285
ResNet-18	+0.007 [+0.001, +0.013]	1.29 [0.23, 2.42]	0.253	0.250
ResNet-50	+0.005 [-0.001, +0.011]	1.16 [0.14, 2.23]	0.255	0.255
ConvNeXt-T	+0.003 [-0.002, +0.007]	0.88 [-0.14, 1.93]	0.231	0.228
ViT-B/16	+0.004 [-0.001, +0.008]	0.79 [-0.25, 1.86]	0.230	0.228

TABLE IV

HAM10000: paired differences in added value between representations, $\hat{\Delta}_a - \hat{\Delta}_b$ (10^{-3} nats, 95% paired lesion-bootstrap CI), fold assignment 0. Exploratory: not corrected for multiplicity.

Comparison	Unadjusted	Source-adjusted
HSV histogram − ResNet-18	14.3 [10.7, 18.0]	10.6 [7.9, 13.7]
HSV histogram − ConvNeXt-T	16.7 [12.6, 20.6]	13.2 [9.6, 16.7]
ResNet-18 − ConvNeXt-T	2.4 [0.7, 4.0]	2.6 [0.9, 4.0]
ResNet-18 − ViT-B/16	1.4 [-0.1, 2.9]	1.5 [-0.1, 3.0]
ResNet-50 − ConvNeXt-T	1.6 [-0.1, 3.2]	1.8 [0.2, 3.4]

indicators, we add a complete-case sensitivity analysis (Section V-D).

Ladder. HSV colour histograms (48-d); ImageNet-pretrained ResNet-18 and ResNet-50 [14], ConvNeXt-T [15], and ViT-B/16 [16]; frozen, pooled features from 224×224 centre crops.

Protocol. 5-fold stratified grouped cross-validation (lesion or patient) with three fold assignments; inner grouped 3-fold CV for the penalty; 2,000 grouped bootstrap resamples. The HAM10000 analysis and the replication were pre-registered, with predictions, hypotheses, and the Holm correction [17] fixed before the respective data were analysed. Deviations and additions, all labelled as such: the complete-case sensitivity analysis (prompted by the missingness audit), the retained fraction ρ, the source-adjusted and paired HAM10000 analyses, the interpretability analyses, and the fine-tuning experiment are exploratory or post hoc. Two implementation errors found during the work (an image-name mismatch that dropped ISIC 2019 images, and a missing-value check that missed NaN) were fixed before any affected result was used.

V. Results

A. Pre-registered analysis on HAM10000

Fig. 1 and Table II show that metadata reduce out-of-fold log-loss by 18.6×10^{-3} nats for colour histograms, with a CI excluding zero in all three fold assignments. For the deep representations the reduction is only 3.7×10^{-3} to 1.7×10^{-3} nats, and the log-loss CIs include zero. Table III expresses the same effect in familiar units: the AUROC gain falls from 0.043 [0.030, 0.056] for colour histograms to 0.007 [0.001, 0.013] for ResNet-18, which remains detectable, and to 0.003 [-0.002, 0.007] for ConvNeXt-T, which does not. Fitting all frozen features and metadata in one model gives nearly the same log-loss as the two-stage design (Table III), so compressing the image to a one-dimensional score does not hide an interaction.

Recoverability of age from the image rises along the ladder (R^2 from 0.06 to 0.39), consistent with Proposition 1 for metadata whose information the image carries. A per-variable analysis did not sharpen this: age, sex, and site retained 14%, 22%, and 18% of their colour-histogram added value at the strongest rungs, and even sex proved recoverable from dermoscopy (AUROC 0.77). Because all three variables became recoverable together, HAM10000 alone cannot isolate the mechanism, which motivates the symptom analysis on PAD-UFES-20 below. Three of four pre-registered predictions held; the predicted non-zero floor was not observed.

B. Shortcut control and paired tests

With image source included in both models, added value still falls (from 15.3×10^{-3} to 2.1×10^{-3} nats). Within single sources, melanoma AUROC rises with capacity, so the deep representations do read lesions and do not merely detect the source. The paired attenuation from ResNet-18 to ConvNeXt-T is 2.4×10^{-3} nats (95% CI $[0.7, 4.0] \times 10^{-3}$).

C. Pre-registered replication

Table V reports the replication. H1 held in 3 of 3 datasets and H2 in 1 of 3 after Holm correction.

At the strongest convolutional rung (ConvNeXt-T), the remaining added value is ISIC 2020: 1.0×10^{-3} nats [0.3, 1.7]; ISIC 2019 (non-HAM): 9.1×10^{-3} nats [3.1, 15.3]; PAD-UFES-20: 206.4×10^{-3} nats [179.9, 232.9]. Attenuation therefore replicates, but how much value

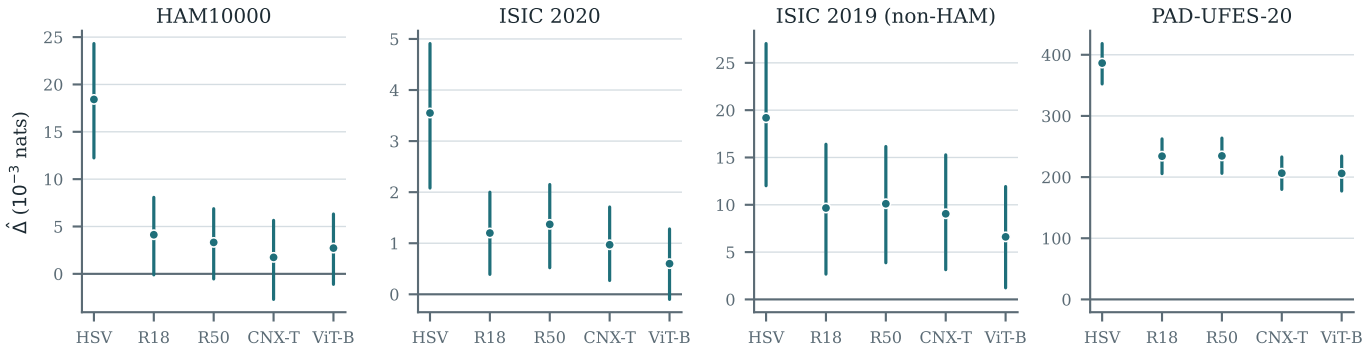


Fig. 1. Added value of patient metadata $\hat{\Delta}$ (out-of-fold log-loss reduction, 10^{-3} nats) along the capacity ladder, with 95% grouped-bootstrap CIs (fold assignment 0). Panels have different y-axis scales: PAD-UFES-20's values are about two orders of magnitude larger. HSV: colour histogram; R18/R50: ResNet-18/50; CNX-T: ConvNeXt-T; ViT-B: ViT-B/16.

TABLE V
Pre-registered replication: added value of all metadata $\hat{\Delta}$ (10^{-3} nats, fold assignment 0) along the ladder, and Holm-adjusted one-sided p-values for H1 ($\Delta_{\text{HSV}} > \Delta_{\text{ConvNeXt}}$) and H2 ($\Delta_{\text{ResNet-18}} > \Delta_{\text{ConvNeXt}}$), grouped paired bootstrap. Family: 6 tests.

Dataset	HSV	ResNet-18	ResNet-50	ConvNeXt-T	ViT-B/16	H1 p_{Holm}	H2 p_{Holm}
ISIC 2020	3.6	1.2	1.4	1.0	0.6	0.003	0.092
ISIC 2019 (non-HAM)	19.2	9.7	10.1	9.1	6.6	0.003	0.241
PAD-UFES-20	386.4	234.2	234.6	206.4	206.1	0.003	0.003

metadata retain at high capacity differs widely between datasets.

Mechanism (PAD-UFES-20). From HSV to ConvNeXt-T, the demographic block retains 0.506 of its added value and the symptom block 0.474. Symptoms are partly recoverable from the photographs (mean AUROC 0.749). Pre-registered M1 did not hold; M2 did not hold. The premise of these hypotheses, that symptoms are invisible in photographs, was false: even within a diagnostic class, ConvNeXt-T features predict the symptoms with AUROC 0.64 to 0.76 (bleeding and elevation are visible), so PAD-UFES-20 contains no auxiliary variable the image cannot partly carry, and the mechanism cannot be isolated with it.

D. Complete-case sensitivity analysis

Restricting every dataset to images with all metadata recorded (Table VII) removes label-informative missingness. Weak-versus-strong attenuation (H1) holds in 4 of 4 datasets; within-deep attenuation (H2) in 2 of 4. On PAD-UFES-20, the standalone information in the metadata falls from 416×10^{-3} to 155×10^{-3} nats, suggesting that much of that dataset's apparent metadata signal reflects how missing values arise, although the complete cases are also a different population. Its retained fraction is nearly unchanged ($\rho = 0.47$), because the leak inflated numerator and denominator alike, but the complete cases (1,089 images, mostly biopsied) form a different population. Complete-case retained fractions are 0.10 (HAM10000), 0.19 (ISIC 2020), 0.50 (ISIC 2019), and 0.47 (PAD-UFES-20).

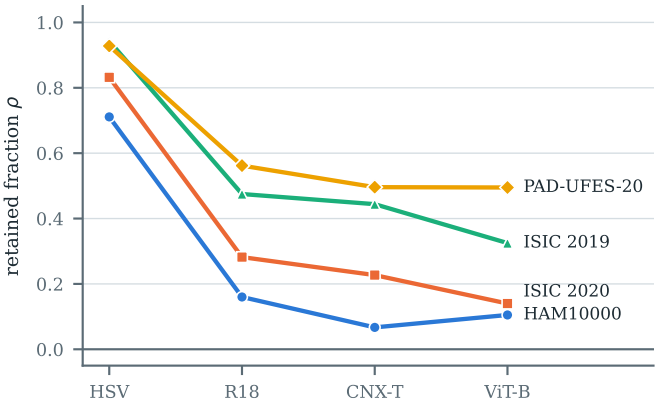


Fig. 2. Retained fraction ρ_k of the metadata's standalone information that is not redundant with image representation k (fold assignment 0; CIs in Table VI). All datasets fall steeply from HSV to the deep rungs, then level off at dataset-specific floors.

E. Fine-tuned image models

[PENDING]

F. Interpretability: where is the evidence?

For each backbone we use the exact class activation map of the out-of-fold head. For ConvNeXt-T, whose pooled features pass through a LayerNorm before the head, the map is exact up to a positive per-image factor: with per-image LayerNorm scale σ and affine weights γ, the logit equals a constant plus $\sigma^{-1}|\Omega|^{-1}\sum_u \sum_j \tilde{w}_j \big(F_j(u) - \bar{F}(u)\big)$ with $\tilde{w}_j = w_j \gamma_j / s_j$ and $\bar{F}(u)$ the channel mean. The overlap statistics are invariant to that factor, and the

TABLE VI

Exploratory: retained fraction $\rho_k = \hat{\Delta}_k / \hat{I}_A$ of the metadata's standalone information that is not redundant with image representation k. $\hat{I}_A$: out-of-fold log-loss reduction of a metadata-only model over prevalence (10^{-3} nats). 95% grouped-bootstrap CIs, fold assignment 0. ρ_k can exceed 1 under synergy.

Dataset	$\hat{I}_A$	HSV	ResNet-18	ConvNeXt-T	ViT-B/16
HAM10000	25.9	0.71 [0.60, 0.80]	0.16 [0.00, 0.27]	0.07 [-0.12, 0.20]	0.10 [-0.03, 0.21]
ISIC 2020	4.3	0.83 [0.72, 0.95]	0.28 [0.15, 0.38]	0.23 [0.08, 0.33]	0.14 [-0.04, 0.25]
ISIC 2019 (non-HAM)	20.4	0.94 [0.84, 1.04]	0.47 [0.25, 0.66]	0.44 [0.23, 0.62]	0.33 [0.12, 0.48]
PAD-UFES-20	416.3	0.93 [0.90, 0.95]	0.56 [0.51, 0.61]	0.50 [0.45, 0.54]	0.49 [0.45, 0.54]

TABLE VII

Complete-case sensitivity analysis: only images with every metadata field recorded (fold assignment 0; one-sided paired grouped-bootstrap p-values, uncorrected). $\hat{I}_A$ and $\hat{\Delta}$ in 10^{-3} nats; ρ at ConvNeXt-T.

Dataset	Kept (dropped)	$\hat{I}_A$	$\hat{\Delta}_{\text{HSV}}$	$\hat{\Delta}_{\text{ConvNeXt}}$	H1 p	H2 p	ρ
HAM10000	9,761 (254)	26.3	18.1	2.6	$\leq$0.001	0.018	0.10
ISIC 2020	32,531 (595)	4.1	3.4	0.8	$\leq$0.001	$\leq$0.001	0.19
ISIC 2019 (non-HAM)	14,156 (1,160)	12.2	13.8	6.1	0.002	0.162	0.50
PAD-UFES-20	1,089 (1,209)	155.4	152.0	73.8	$\leq$0.001	0.520	0.47

TABLE VIII

Class-activation evidence vs. expert lesion masks (HAM10000, ResNet-50, 2225 images). Enrichment: share of positive evidence inside the lesion divided by the lesion's area share. Significance at $p \leq 0.05$.

Evidence map	Enrichment > 1	Shift null	Pixel perm.
Predicted class	32%	10%	30%
Melanoma (all images)	84%	37%	82%

TABLE IX

Melanoma evidence (exact CAM) vs. expert lesion masks across backbones (HAM10000; all melanomas plus a matched sample of other lesions). Significance at $p \leq 0.05$: exact circular-shift null vs. pixel permutation. Last column: median lesion area fraction of detected / missed melanomas.

Backbone	Images	Enrich.> 1	Shift null	Pixel perm.	Area det./miss.
ResNet-18	2,222	90%	53%	89%	0.49 / 0.37
ResNet-50	2,222	84%	36%	82%	0.52 / 0.37
ConvNeXt-T	2,222	91%	56%	90%	0.49 / 0.33

reconstructed logits matched the head's logits with correlation 1.000 for every backbone.

Mapping the predicted class's evidence produced an apparent link between errors and off-lesion attention ($p < 10^{-49}$; Table VIII). This is an artefact: for "not melanoma" predictions, the evidence for that class lies in the surrounding skin. With the melanoma-evidence map used for every image (Table IX), melanoma evidence is concentrated on the lesion in 84%–91% of images across the three backbones. Missed melanomas show no less on-lesion evidence; instead they are consistently smaller lesions ($p \leq 5 \times 10^{-12}$ for every backbone). Pixel permutation declares 82%–90% of maps significant, against 36%–56% under the exact shift null, consistent with the inflation predicted by its disregard of spatial autocorrelation [10], [11].

VI. Discussion and Limitations

What the results mean. On every dataset, metadata add clearly more to a colour-histogram model than to frozen deep representations (H1 held in all three pre-registered replications). Attenuation within deep representations is small and inconsistent: H2 held only on PAD-UFES-20. How much value remains at high capacity depends on the dataset rather than on the imaging modality (Table VI). A ConvNeXt-T representation leaves $\rho = 0.07$ of the metadata's standalone information non-redundant on HAM10000 and $\rho = 0.23$ on ISIC 2020, but $\rho = 0.44$ on ISIC 2019 (non-HAM, largely BCN20000, which deliberately includes hard-to-diagnose sites [5]) and $\rho = 0.50$ on PAD-UFES-20; the complete-case values are similar (0.10, 0.19, 0.50, 0.47; Table VII). The frequently reported benefit of metadata [1], [2] is therefore partly a statement about weak image representations, and the size of the part that is not differs severalfold between datasets.

Practical implication. Metadata fusion should be evaluated against the strongest available image model, and the retained fraction ρ reported alongside the gain, so that a benefit that merely compensates for a weak backbone is not mistaken for complementary information.

Limitations. (i) Only frozen representations are compared along the ladder; fine-tuning is examined separately (Section V-E). (ii) $\hat{\Delta}$ estimates the reduction achievable by a logistic second stage, not $I(Y; A \mid S)$ itself. (iii) Capacity is ordered by image-only performance; the ladder is not a controlled intervention on a single architectural factor. (iv) The mechanism suggested by Proposition 1 is not isolated: the pre-registered symptom test failed because symptoms are partly visible even within a diagnostic class. (v) ISIC and PAD-UFES-20 images come from public mirrors; official metadata were used, and counts and contents were checked against the published descriptions. ISIC 2019

images without a lesion identifier were grouped individually. (vi) Metadata missingness is label-informative in all four datasets, severely so in PAD-UFES-20. Complete-case analysis removes this leak but analyses a selected subsample. (vii) All analyses are retrospective and internal (cross-validated); nothing here is a clinical claim. (viii) Saliency overlap is evidence of spatial correspondence, not of causal reliance [18].

VII. Conclusion

The value of patient metadata to a skin lesion classifier is not fixed: it depends on what the image model already extracts. Proposition 1 makes this precise. Grouped, pre-registered experiments show that metadata add far less to deep representations than to colour histograms on HAM10000, and this weak-versus-strong attenuation replicated in all three independent datasets. Attenuation among deep representations did not replicate consistently, and the share of metadata information left non-redundant by a strong representation ranged severalfold across datasets. Claims that metadata improve a classifier should therefore be made against strong image baselines and accompanied by the retained fraction, not a gain over a weak model.

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